<u>Ch-28 (Investment strategy)</u>

Why is Investment strategy important?

- Competition with other life insurance companies and financial institutions.
- Potential customers of WP and UL PH will be influenced by the recent return on the prod.
- Profitability of company
- Security to PH because able to meet the benefits and meet expectations.

<u>Asset characteristics</u>

- Return, security, Marketability, expense, term and currency. (TERMSC)
- Statuary constraints.
- Most important is return for which we should think about: how much is expected return? Is the return real or nominal? How is the return split between income and capital redemption? Is the yield sufficient to meet investor's expectations? Variance of the return? And how is the return affected by tax because tax reduces the return, tax regimes may favour some assets or/and may favour income over capital gains?

Type of assets

	Govt fixed interest bond	**Govt index linked**	**Corp fixed interest bond**
Terms	Depends on market, generally 0-20	Same as fixed bond	Smaller compared to govt
expense	low	Same as fixed bond	high
return	Nominal but fixed Coupon similar to market yield zero coupon bond higher redemption value	Same as underlying index	Better than govt, value of bond will fluctuate with market but won't matter if help till maturity

Marketability	High	Amounts and variety are smaller, thus lower	less
security	Very but less than cash	very	Depends on credit rating of company
Currency	local	local	Any

- Equities
    - Offers dividend which is likely to increase in real terms.
    - MV- increase but it is volatile which is a problem for long-term income because it will not help in improving the solvency position - less than purchased value.
    - Risk attached to income and capital value.
    - Companies may go bankrupt or perform badly.
    - In some countries equities may not really be an option because of the size and reliability of the local market.
- Properties
    - Term - Very long
    - Expense - Significant dealing expense, incurred even in just administering the asset. Large investment and thus problem of diversification for small life insurance companies.
    - Return - High return. Income in the form of rent, MV can vary significantly from economic cycle.
    - Marketability - Unmarketable
    - Secure investment however actual income stream may suffer occasional interruptions.
- Cash – Earns spot rate of interest, most secure – highly liquid, no dealing cost and lowest return

- Comparative return : equities and properties>corp bond>govt bond>cash

<u>The principle of investment</u>

Can be stated as:

- Select investment strategy which matches with liabilities in terms nature, term and currency.
- *Maximize overall return (investment income and capital gain) on the assets.*
- The extent of deviation from matching will depend on free assets and risk appetite, i.e., risk level company can absorb to keep it functioning at best of its capacity.
- *The company should consider the ESG issues as well*

<u>Asset liability mismatching requirements</u>

- Assets proceeds identical to liabilities, as they occur, resulting in no investment risk.
- However, it is impossible in practice and eliminates all the probabilities of making profit.
- Desirable if a company has very low free assets and otherwise the probability of ruin will be unacceptably high.
- Nature, terms and currency of the liabilities must be considered
- Liability outgo = claims + expense + tax + increase in reserve – premium- inv income

Nature

- Guaranteed – Amount payable is fixed.
- Index linked – guarantees whose amount is linked to price, earnings, inflation such index. Eg – expense payments linked with inflation
- Investment linked – amount is determined by the value of the underlying asset. UL
- Discretionary – payments at discretion of LI company, so future bonus payments of WP.

<u>Guaranteed in terms of money-</u>

- The company's goal is to meet its guaranteed liabilities.
- This means investing in assets with proceeds the same as liability outgo.
- Terms of the liability outgo will also need to be considered, along with the probability of payment being made.
- In practice it is impossible to find assets that exactly match our liability outgo.

- At early stages of the contract, if the term of liability is smaller than the assets, proceeds from assets may exceed the net liability outgo. The excess will have to be reinvested at the unknown rates, making strict matching impossible and introducing reinvestment risk.
- Therefore, the best match is what we can hope for.
- The technique called immunization may be used, where companies would be protected against assets not covering the liabilities in the event of interest rates moving, provided the movement was not too large. But this technique is subject to many practical problems. The problems and limitations are:
    - It immunizes against both profits and losses
    - It can be difficult to immunize against real liabilities
    - The theory only works with small interest rate fluctuations
    - It assumes flat yield curve
    - analyze the situation every day and change investments accordingly
    - *Assets of the required term may not exist*
    - The timing of assets proceeds might not be known, and that of liabilities can only be estimated.

<u>Guarantees in terms of price index or similar</u>

- A suitable match - securities linked to the same index and also match\ expected term of the liability outgo.
- *Substitute - assets that are expected to provide a 'real' return.*

1. Discuss how equities would be a match for liabilities linked to price index?

Not a perfect match because their return is not guaranteed.

However, to some extent, the return is in line with price inflation, so a reasonable match.

Suitability of a match depends on how long a term period is considered.

Short period – poor match – high volatility.

Longer period – better match – less volatile

<u>Discretionary</u>

- Non-guaranteed portion of WP liabilities or surrender value when it is not guaranteed.

- Main aim – maximize the discretionary benefit and meet SH and PH expectations.
- PH expectations may restrict the investment freedom thus, investing in assets with higher return with medium risk almost surpassing the inflation
- Equities of established companies.
- The decision on assets for discretionary benefits will depend on several factors:
    - The bonus philosophy of the company – a company which gives low guaranteed benefits and concentrates on terminal bonus or a significant terminal dividend may tend towards investing in equities.
    - The level of free assets – that can support a riskier investment strategy.
    - Risk appetite of company
    - PH expectations will be influenced heavily by the investment strategy company publishes, and so should not deviate too much from publicized risk vs return stance.
    - The views of the company to the relative performance of the asset classes concerned

<u>Investment linked</u>

- UL
- Value of asset in accordance with a definite formula, based on the value of a specified fund of assets.
- Companies can avoid the matching problems by investing in the same assets used to determine the benefits.
- But sometimes company do not invest in the same assets because they thought that it could profit from such a deliberate mismatch

Term

- Estimate liability outgo in each future time period.
- Match the liabilities with assets with the same timing and amounts of payment.
- Difficult to find exactly matching assets therefore an approximate approach might be used.
- For example,assets with the same Discounted mean term (DMT) as liabilities.
- *DMT is defined as the weighted sum of the terms of the payments, where the weight attributed to each term is the PV of the payment at those terms*.

- DMT=∑ti*expected cash flowwi*vi∑expected cashflow*vi where viis the discounted factor
- If assets and liabilities will have identical DMT, then our investment will move the same as our liability in the event of interest rate movements.

Currency

- Liabilities denominated in a particular currency should be matched by assets in the same currency to reduce the currency risk.
- An alternative way is to invest in overseas assets and hedge the currency risk by using appropriate currency options.

1. Why might companies want to invest overseas?

- If some liabilities are denominated in that currency.
- Increase diversification, or invest in assets not available locally or some countries' assets might be giving good value compared to the home country.
- Only possible to do so, its company allocates adequate free assets to cover for potential downside risk.

Risk of cashflow mismatching:

Risk	**Solution**
Falls short	Project cash flows at different scenarios
SCR	Analyse at different assumption and keep more reserves if required

Free assets

- Due to the free assets of the company, it can deviate from the matching strategy to improve overall return on its assets and thereby benefit its PH, through higher bonuses or lower premium rates and SH through higher dividends.
- Assets with higher expected return = higher variance of return
- Free assets are used as a cushion to reduce the risk of insolvency.
- So greater the free assets,
    - greater the freedom in choosing investments for guaranteed liabilities.

- A riskier profile of assets can be chosen by the company for discretionary benefits.
- For investment linked higher return can be expected with appropriate mismatch.

Developing an appropriate investment strategy

- Starting point will be to identify the strategy to match the liabilities.
- Matching in terms of the nature, term and currency of liabilities and split them in categories of nature
- Start with investment linked and match these exactly using the underlying benefit formula.
- Match the index linked liabilities with reference to an index – if not possible then to the nearest thing.
- Match the guaranteed liabilities in money terms using government bonds
- For discretionary liabilities, it is ideal to invest in heavily in medium risk equities and some properties due to illiquidity risk.
- Modify the above strategy to include sufficient cash for the company to operate on a daily basis without needing to realize any non-cash assets
- Now the company has reasonable base of investment strategy representing 'minimum risk' and thus can consider divergences for greatest expected reward to both SH and PH while maintaining sufficient risk control commensurate with the company's risk appetite and resources.

Model office to determine an investment strategy /determining appropriate divergence from base strategy / ALM

- Determine the amount of free assets which can be used as cushion because
    - SH may not want to use all their assets as 'security' for the fund.
    - SH may also think that giving PH max return does not justify the risk of their free assets to disappear
    - Regulator may also restrict the use of certain free assets if regulations specify that they are inadmissible to cover reserves

Asset liability modelling

- *The liabilities and assets would then be projected forward on assumptions that represent expected future experience and variation from these*

- *For the assets, stochastic models can be incorporated to project investment income in future and changes in capital values. Inflation rate model can be used to project the future expenses on the liabilities side*
- *The projected liabilities and assets can then be valued at the end of each year to ensure the company is always solvent.*
- *After performing the asset/liability model projections of the company's future assets, compare these total assets against the reserves (liability)*
- *Check the excess of these assets over the liabilities exceeds any minimum capital requirement for the entire projection period for most of the model runs*
- *Calculate some measure of aggregate profitability*
- *Repeat these last steps assuming a different investment strategy until the target profitability of insolvency is achieved.*
- *Choose the strategy with equal solvency risk and highest profitability*

Three potential disadvantages of these strategies are-

- Modelling risk – the conclusion of ALM will be valid to the extent of the model and the inherent parameters are valid.
- Portion of free assets - Although the owners of the free assets might justifiably not want all of those assets to be put at risk in supporting an unmatched investment strategy, that does not imply that the model should necessarily exclude certain assets. For instance, we can still have a constraint along the lines of 'we accept that 50% of free assets will disappear with a probability of 1%' but model the entire asset portfolio and interpret the results accordingly. This could give more useful results.
- If SCR is not constant then an approach which is less vulnerable to error would be included in the ALM to ensure that it is modelled correctly.

So there are three decision variables here –

- The riskiness of investment strategy
- Level of Free assets
- Probability of insolvency

The investment strategy can become riskier if

- By reducing the diversification

- For assets backing some given liabilities reduce the amount of matching by – nature, currency or term
- Looking at assets in isolation increases the duration
- In both cases –
    - Move from govt bond to corporate bond
    - Move from AAA rating to lower
    - For equity move from established company to new company
    - Or established sector to a new sector
- For a proprietary company, the process can be extended further and the effect of the investment strategy on future shareholder earnings can be considered. In particular, an investment strategy can be determined that maximizes shareholder income whilst keeping the risk of insolvency sufficiently low, bearing in mind the available level of free assets.

Reinsurance

- Insurance of insurance
- It is an arrangement whereby one party (the reinsurer), in consideration for a premium, agrees to indemnify another party (the cedant), against part or all of the liability assumed by the cedant under one or more insurance policies, or under or more reinsurance contract.

Type of reinsurance

- Proportional reinsurance
- Non-proportional reinsurance
- Financial reinsurance
- Facultative and obligatory reinsurance
- Coinsurance

Proportional reinsurance

- Cedant and reinsurer share premiums, claims and liabilities in predetermined proportions of each policy
- Eg - if the reinsurer agrees to take on 40% of the risk on a contract, they receive 40% of the premium and are contractually obligated to pay 40% of any claim that arises on that policy.

- They are of two types –
    - **Quota share** – a specified %age of each policy is reinsured
    - **Individual surplus** –
        - Every risk has to be ceded,
        - Different % ceded for different risk
        - Reinsured amount is the excess of the original benefit over the cedant's retention limit on any individual life
- Company will likely adopt mixture of both, i.e., retaining a % of each policy up to a maximum limit.

Basis of premiums–

- Original terms (coinsurance) reinsurer
    - The cedant will provide the reinsurer with the premium rates it is using for the class of business it wishes to reinsure so that reinsurer can check that the premium rates charged by insurer are correct
    - Then the reinsurer will determine the rates of reinsurance commission it is prepared to pay to the cedant for the business.
    - In doing so, it will have to know likely future experience and its knowledge of the quality of cedant's underwriting.
    - It will also want to make a profit on the business
    - Premium received by insurer will be split in same %age as claims.
- Risk premium reinsurance
    - Independent of premium charged by insurer to PH
    - A specified premium is charged for a risk, which may or may not vary over the term of the policy.
    - Reinsures the part of sum assured or sum at risk, i.e., excess of benefits payable over the reserves.
    - determines by assessing the likely experience or probability of claim and then adding expense and profit margins*reinsurer's portion of SAa, Risk premium = qx+tRBt
    - Risk premium can be changed because.

- If reinsurance is based on Sum at risk, then sum at risk would change over time, leading to change in the reinsured amount at risk.
- Risk premium rate will change as it will be based on the age of PH.
- If pricing basis is not guaranteed, then change due to pricing review.

Non-Proportional reinsurer

- Claim cost is shared based on the size of the claim.
- Under this cedant agrees to pay all the claims up to a predefined limit, called retention limit and beyond that remaining of the claim amount is being paid by the reinsurer.
- It could be required when the insurance company runs into trouble by receiving more claims than expected, that is when this type of reinsurance will be beneficial.

Excess of loss reinsurance

- Reinsurers pay any claim of an individual risk in excess of a predetermined retention limit.
- It can be enacted on a risk basis or on occurrence (when aggregate loss from any occurrence of an event exceeds the predetermined retention).
- The premium base will include margins to reflect the uncertainty of the reinsurer's total claim amount.

Catastrophe XOL

- The aim of catastrophe reinsurance is to reduce the potential loss to the cedant due to any non-independence of the risks insured.
- This is mainly beneficial for group insurance because the employees insured in the group insurance are working in the same environment. Some accidents, such as an explosion or industrial disaster or crash, would probably affect a large number of these employees.
- Yearly basis cover and renegotiated yearly.
- Beneficial for reinsurers if it feels like risk posed by insurers has been changed.
- Cedant will also have freedom to check out other options.
- There is no standard definition of what is catastrophe, but typically to qualify there needs to be a minimum number, say five, deaths from a single incident within a specified time.
- Treaty will include how much reinsurance company will pay, i.e., retention amount to maximum amount reinsurer will pay. Beyond that claim amount will go back to reinsurer.

Stop loss reinsurance

- Reinsurer pays the aggregate net loss over the determined retention for a portfolio over a given period, usually a year.
- This caps the cedant loss in any period

Financial reinsurance (Fin Re)

- Improve the accounting and supervisory solvency position of the cedant, i.e., aims to manage the insurer's capital position because if taken a loan then it would have been added in liabilities and in assets
- It involves small transfer of reinsurance risk from the cedant to the reinsurer.
- Insurer needs money – reinsurer pays initial payment to insurer as commission related to business reinsured, insurer pays back to reinsurer by adding repayments in premium.
- Another method of repayment involves contingent loans – where repayment is contingent on future profits.
- Cedant doesn't require to keep the reserve for the repayments.
- In this assets will be increased by the amount of loan and no need to show the liabilities as these will only materialize in future if business is profitable.

Facultative and obligatory reinsurance

Facultative – freedom of action

Obligatory mean – obligation

- There can be four cases of this type of reinsurance

  Facultative/ Obligatory, Facultative/ Facultative, Obligatory/ Facultative, Obligatory/ The treaty is a legal contract governed by contract law, which will include international trade and shipping agreements and a large body of case law

Coinsurance with insurer

- Risks are different to each other and thus diversified
- Less overlap
- Share risks

Reasons for reinsurance

Uses (FAILRRAT)

- **F**inance of new business strain
    - Increase in available capital or through a reduction in its financial requirement.
- **A**llow aggregation of risks that the cedant cannot manage on its own, so allowing manufacturing of product lines'.
- **I**ncrease profit
    - Reinsurer can write business more cheaply because of different capital requirements, taxations and assessment of risk
    - The risk diversification will be more than any individual cedant, in terms of number of individuals covered, geographical area, original sales channel.
    - This decreases the random fluctuation risk and the quality of data reduces the parameter risk.
- **L**imit the claim payout.
- <u>**R**educe insurance parameter risk –</u>
    - There is a risk that the level of claims may be different from expected. This may be due to incorrect pricing, underwriting failures, fraudulent activities, etc.
    - If the A>E => loss from parameter risk.
    - Quota share could be used and reinsurer will review the premium rates and the insurer's control because he won't be happy to pick the cost of serious failings in the insurance company's pricing and controls.
- **R**educe claim payout fluctuations
    - Helpful when = small number of contracts with very high claim amount or the lives insured are not independent
    - For high small number of contracts – proportional individuals surplus reinsurance
    - For the lives that are not independent – non-proportional reinsurance
    - Large fluctuations = insolvency risk, undesirable level of free assets, fluctuations in dividends to SH
- **A**void large single loss

- **T**echnical assistance
    - Reinsurer has a considerable degree of expertise in underwriting, product design, pricing and systems design.
    - Important when launching new prod
    - Existing lines -> underwriting
    - Since reinsurer has more data they can appropriately provide insight
    - Quota reinsurance

<u>Considerations before reinsuring (LACCT)</u>

<u>Legal risk</u>

- Reinsurance is governed by treaty
- There are usually many clauses to be negotiated to cover numerous contingencies and risks, each one potentially impacting the price of the reinsurance.
- Due to the operational risks involved and the impact of disputes where contracts have not been finalised, regulators around the world are increasingly focused on ensuring that reinsurance treaties are complete and signed.

<u>Amount of reinsurance : Retention limits</u>

General factors to consider while setting the retention limit

- Risk appetite
- Free assets
- Effect of limit on regulatory capital requirement
- Level of familiarity with underwriting process
- Average claim amount and its expected distribution
- Terms of reinsurance and their impact on limit
- Impact on profits
- increase of sum assured in future
- Company’s retention on other products

Approaches to determine retention limit

- Stochastic stimulations – reinsurance only
    - Define the safety target for the company –
        - Probability of insolvency at certain level or
        - By defining the probability that the loss does not exceed a proportion of the company earnings.
    - Test the retention limit by picking a trial limit.
    - Run the stimulations on different scenarios.
    - Check the probability of insolvency, and see if the limit is safe and acceptable. Otherwise use different limits.
- Stochastics stimulation – reinsurance with fluctuations in reserve
    - Consider the cost of mortality fluctuations reserve and cost of obtaining reinsurance.
    - Procedure

Cost of obtaining reinsurance:

- Decide claim volatility criteria beyond which company cant go.
- For differing retention limit, model the function
    - Total claim net of reinsurance – total risk premiums net of reinsured risk premium
    - Net loss, X = [C(g)-C(r)]-[P(g)-P(r)]
    - Then P[X>Retention limit]=0.01
- Choose the retention limit that will satisfy this result.
- And then check whether the protection can be done more cheaply using a mortality fluctuation reserve.
- To do this, assume the cost of risk premium reinsurance to be spent on mortality fluctuations reserve. The cost of holding a reserve is M*(j-i) where j is the return SH expected return and i is the return earned.
- Say 60% of reinsurance risk premium is used mortality fluctuation reserve, and i=7% and j=9%,
    - Then M=0.6*reinsurance premium/(0.09-0.07)=30P(r)

    - Now companies only have 40% of reinsurance risk premium to buy reinsurance, so retention limit would need to be increased.
    - X = [C(g)-C'(r)]-[P(g)-P(r)]-M =[C(g)-C'(r)]-[P(g)+29P(r)]
- Compare the protection offered under this new construction against the offered earlier, recalculate P(X>Retention limit], if < 1% previously obtained, then using a mortality fluctuation reserve is cheaper than using reinsurance, and would therefore be a preferred strategy.

Cost

- Reinsurers would want to make profit and meet their costs of capital and expense - reducing the absolute level of profit.
- More the company reinsures, lower the risk of fluctuations but lower the expected return.
- However, the risk reduction caused by reinsurance may leverage up the return.

Counterparty risk

Can be countered by

- Deposits back
    - In some countries, regulator requires the reinsurer to collateralize or 'deposit back' its share of the total reserve under a reinsured contract with the cedant, i.e.,
    - Customer Pays Premium to Primary Insurer and Reinsurer takes risk & earns profit, BUT, Reserve Money Stays Locked in Cedant's Vault As Deposited Back or Collateral.
    - Even where it is not a supervisory requirement, deposits back are sometimes done so that the cedant gets the benefit of reinsurance, whilst at the same time being able to maintain a reserve for the whole contract and hence maximise the funds it has to invest.

Type of reinsurance and the way in which amount is specified will depend on:

- The reason for using reinsurance.
- The reinsurance cost
- The type of business
- Legal condition applying
- The form of reinsurance coverage in the market

Strategy to decide how much reinsurance to use

- Optimal strategy on the basis to achieve the desire level risk with minimum cost (this is equivalent to achieving max returns for a required level of risk target)
- Stochastic model office will be used → mean mortality and random mortality fluctuations will be modelled stochastically as random variables
- Sensitivity of results → to parameters, and model points (e.g. assumed distribution of sum assured in the model points)

<u>Chapter 26 (Underwriting)</u>

- Classify and identify different levels of risk of PH in respect to the future experience and ensure they are charged fairly together with terms and conditions of cover.
- Helps to control the quality of lives accepted for insurance.
- It aligns the risks written by a company with the pricing performed by the actuary.

<u>Why is underwriting important? Or uses of underwriting? Or how underwriting helps in managing risk?</u>

- <u>Protect life insurance companies from **anti-selection.**</u>
- PH knows their health better, the insurer might not be able to identify that. Resulting in a pool of sub-standard lives and high mortality experience. This can be minimized by the Underwriting approach and premium risks.
- Underwriting approaches should be in line with the market.
- <u>Identify lives substandard health risk</u> for whom either
    - special terms would need to be quoted - These special terms would be – higher premium - same sum assured, same premium and lower SA, reduced terms, or exclusions, i.e., with certain terms and conditions defining what is covered and what is not and the circumstances.

Exclusion clause –

- least preferred option since cover could be required for protection
- Causing adverse publicity
- Very difficult to enforce at the time of claim.

- Official recorded cause of death can be vague or ambiguous or misleading in order not to invalidate the insurance claim.
- Hence exclusion clauses are unlikely to be effective
- Or decline the policy
- Or defer the decision to offer based on current health but impose a waiting period and observe the conditions, reassess the risk after some time and consider reassigning or asking for more medical checkups.
- Or offer an alternative policy.
- It helps to ensure that <u>all risks are rated fairly</u>.
- ensure that <u>actual mortality experience does not depart too far from that assumed</u> in the pricing of the contracts being sold.
- Risk of over insurance can be countered by financial underwritings.

Importance of underwriting for each product

Product	Underwriting is
Term assurance	Critical
Whole life assurance	Very important
Endowment	Important if regular premium, not very important for single premium
Pure endowment	Irrelevant
Immediate annuity	Not important. Unless impaired life annuities

Types of underwritings

- Medical
- Lifestyle – the influence of sporting and hazardous leisure pursuits on the risk, and the extent to which the lifestyle might increase the possibility of dangerous diseases
- Financial
- Claims underwriting

<u>Process</u>

<u>Medical Underwriting</u>

- Obtain evidence about the health of the applicant to assess state of health as standard or sub-standard and accordingly offer terms
- The aim here is to improve the marketability of the contract and increase fairness to controlling risk.

Medical evidence can be obtained from :

- Questions on the <u>proposal form</u> completed by the applicant, such as -
    - Height and weight,
    - Smoking and drinking habits
    - Current health and details of any current treatment being received
    - Personal and Family medical history
    - Occupation and potentially dangerous pastimes

The extent to which these answers are reliable depends on the culture of the country and on the distribution method.

Ideally compiling the proposal form will be the only form of underwriting done. This is very cheap, however the only problem is what to do if some of the answers suggest non-standard risk, or if sum assured is so high that the company wants to be more certain about the risk involved.

<u>The limit on sum assured, above which some real medical evidence is required, is known as 'medical limits'</u>. This will reflect the balance between the cost of obtaining the medical evidence and the cost of the risk of not having it.

- Reports form medical doctor that the applicant has consulted –
    - However, there could be privacy of records or data protection
- A medical examination carried out on the applicant (Basic Blood and urine test)–
    - This will be a basic half-hour examination covering many of the points in the proposal form but allowing the doctor to check any of the obvious problems.
    - It is a costly option.
    - There is also a risk of discouraging healthy applicants if resorted to too much.

    - It is therefore only used in the event of unsatisfactory answers on the proposal form, or a high sum insured or if no reports are available from previously consulted doctors.
- Specialist medical test on the applicants (Xrays, blood test, electrocardiograms)
    - This the most expensive and intensive option
    - Some of these tests might also be automatically required for very high sums insured on temporary life assurance contract.

Since all these options add cost to the insurance company, therefore, the extent to which they are used in a particular case depends on the extent of the loss that the life insurance company will make if it mis-estimates the state of health of the applicant.

Extensive underwriting can make the company lose the business therefore, market practice also needs to be considered.

Other evidence

- Occupation
- Leisure pursuits
- Normal country of residence (climate, prevalence of disease, availability and quality of medical facilities, level of violent crime)

Financial underwriting

To ensure that:

- Premium is affordable to applicant – to control the persistency risk.
- The person is not trying to commit fraud. An indication of this could be having higher levels of sum assured than could be justified by the applicant's current circumstances.
- For term assurance, the ratio of sum assured to the salary of applicant can be considered.
- Company needs to ensure that policy benefits is in the line of financial needs and lifestyle of such a person.

Claims underwriting

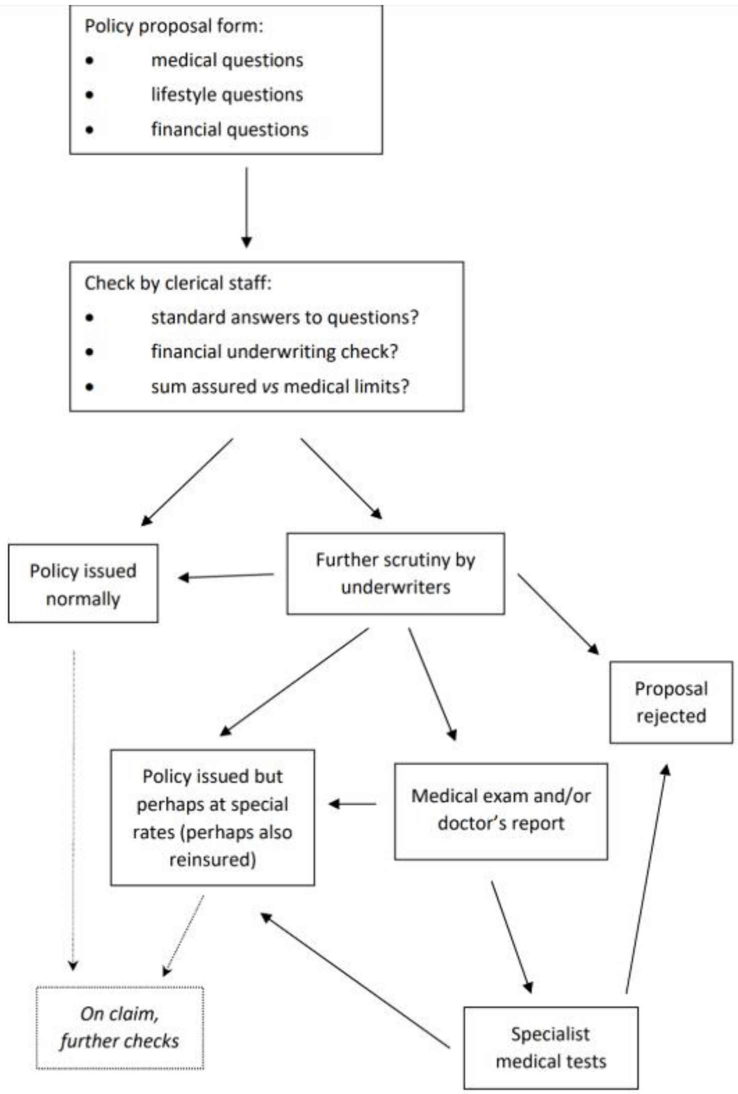
- A life insurance company may also decide to 'underwrite' at the claims stage
- Reject claim 0 Suspicion of non-disclosure or fake claim or subject to exclusion clause

Interpretation of the evidence

- The evidence needs to be interpreted of the standard of health required
- Proposal form will be reviewed by clerical staff. They will look through the answers, converting the numerical fields into ok and not okay.
- This will be done by specialist underwriters who will make use of:
    - Any Doctors are employed by the life insurance company for this purpose.
    - Underwriting manuals prepared initially or by the major reinsurance companies.
- For basic medical evidence and disclosures on an application form, specialist underwriting software is proving increasingly efficient at assessing information and making rating decisions.
- For moderately complex and financial underwriting cases, experienced professional underwriters are required.

Specification of terms

- Applicants who fail the financial underwriting test would be declined.
- Before declining, the insurer may ask for additional information eg – an explanation of the need for the large sum assured.
- However if an attempt on fraud is suspected then the insurer may decide to decline the proposal as a small price to pay for reducing the risk of fraud and avoiding the costs of further investigations.
- Applicants whose state of health reaches the required standard can be offered the life insurance company’s normal terms for the contract.
- Other applicants will be offered Special terms, unless their state of health is such that the company will not accept them on any terms, will be declined at least temporarily.
- The company will normally rely on the reinsurers’ and industry data in rating normal lives, since the company’s own experience for any particular experience will normally be small.



__Determining the level of underwriting to use__

- The company needs to ensure that the increased volumes that result from more relaxed underwriting outweigh the cost of increased anti-selection and any increased costs of obtaining reinsurance.
- The main goal is – increased profits, reduced underwriting expense and increased attractiveness of the prod
- Conversely, companies would want to improve mortality experience by avoiding any non-standard risk, however relative cost of high level of underwriting will be more particularly for small sums assured.
- The two cost is doing so:
    - The actual expenses of the underwriting

- The discouragement of new business

- In a way underwriting is justified to the extent it pays for itself in improved experience

Factors need to be considered –

- The expense associated. Underwriting involves a number of costs, such as salary and medical reports, which will increase the overall costs to the company.
- The extent and the financial significance of anti-selection risk.
- The interaction between the level of underwriting and the potential level of sales.
- Claims underwriting may deter people from taking up a contract due to the uncertainty as to whether a claim may be accepted.
- The effectiveness of the proposed underwriting
- The more detailed the underwriting, greater the homogenization of achieved risk.
- The impact of regulation
- Interaction between underwriting and terms as in margins offered by the reinsurer.
    - Requires a certain level of underwriting in place and will carry out reviews.
    - The terms offered will depend on the past claim experience of the insurer and the underwriting standards in place
    - This involves potentially considerable operational risk, as the reinsurer may not pay claims if the underwriting was not up to the specified standard
- How to vary underwriting criteria by age, sum assured, target market and various other factors.
- Type of product, Distribution channel, competitiveness, Operational risk
- Marketing risk
    - Underwriting is a barrier in sales as well as time-consuming and invasive.
    - Less or no underwriting is good for marketing risk could be higher for such policies but overall cost may not be, making pricing competitive.